

Predicting Open-Ended Response

Group 15

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Question

What predicts having any likes/dislikes for either party? Who are we including/excluding in the open-ended sample?

A respondent is **excluded** from the open-ended sample if either

- they answered “No” to the like/dislike screener (`*_binary == 5`), **or**
- they said “Yes” but provided no actual open-ended text (`*_binary == 1` *and* the text field is `NA`).

1 Data prep

```
setwd("/Users/yzhang/Desktop/r_projects/mdm_final")
rm(list = ls())
pacman::p_load(glmnet, coefplot, tidyverse, sandwich, lmtest)

data <- read.csv("data.csv", header = TRUE, na.strings = c("", "NA"))
dim(data)
```

1.1 Recode the four outcomes

For each like/dislike pair, the new 0/1 outcome is:

- 1 if *_binary == 1 AND the text field is non-empty (gave usable text)
- 0 if *_binary == 5 OR (*_binary == 1 AND text is NA)
- NA if *_binary itself is NA (drop these rows later)

```
recode_y <- function(bin, text) {
  text_present <- !is.na(text) & nchar(trimws(text)) > 0
  y <- rep(NA_integer_, length(bin))
  y[bin == 5] <- 0L
  y[bin == 1 & !text_present] <- 0L
  y[bin == 1 & text_present] <- 1L
  y
}

data$y_dem_like <- recode_y(data$dem_like_binary, data$dem_like)
data$y_dem_dislike <- recode_y(data$dem_dislike_binary, data$dem_dislike)
data$y_rep_like <- recode_y(data$rep_like_binary, data$rep_like)
data$y_rep_dislike <- recode_y(data$rep_dislike_binary, data$rep_dislike)

supply(data[, c("y_dem_like", "y_dem_dislike", "y_rep_like", "y_rep_dislike")],
  table, useNA = "ifany")
```

1.2 Drop columns that should never be predictors

We drop:

- IDs and the survey weight (not substantive predictors).
- The raw outcome components.
- age_cohort (collinear with age).
- Items not asked every year (understand_issue, too_complicated, power_matters, dem_satis) and items with high missingness (dem_liking_cses, rep_liking_cses, therm_lib, therm_consv). Keeping them would drop the analytic sample below 15,000 after na.omit.

```
drop_cols <- c("caseid", "uniq_id", "weight",
  "dem_like_binary", "dem_dislike_binary",
  "rep_like_binary", "rep_dislike_binary",
  "dem_like", "dem_dislike", "rep_like", "rep_dislike",
```

```

      "age_cohort",
      "understand_issue", "too_complicated",
      "power_matters", "dem_satis",
      "dem_liking_cses", "rep_liking_cses",
      "therm_lib", "therm_consv")
data <- data[, !(names(data) %in% drop_cols)]

```

1.3 Collapse multi-level ordinals

Per-level coefficients on 7-point scales are hard to read in a table, so we collapse them. Out-of-range codes (DK / RF / 0 / 9) fall through to NA and get dropped by `na.omit` later. The middle category is set as the reference so coefficients read as “vs moderates / independents.”

```

collapse_7pt <- function(x, labels) {
  factor(dplyr::case_when(
    x %in% 1:3 ~ labels[1],
    x == 4 ~ labels[2],
    x %in% 5:7 ~ labels[3],
    TRUE ~ NA_character_
  ), levels = c(labels[2], labels[1], labels[3])) # middle first = reference
}

data$pid7 <- collapse_7pt(data$pid7, c("Dem", "Ind", "Rep"))
data$ideology <- collapse_7pt(data$ideology, c("Lib", "Mod", "Con"))
data$dem_ideo <- collapse_7pt(data$dem_ideo, c("Lib", "Mod", "Con"))
data$rep_ideo <- collapse_7pt(data$rep_ideo, c("Lib", "Mod", "Con"))

data$educ7 <- factor(dplyr::case_when(
  data$educ7 %in% 1:2 ~ "NoHS",
  data$educ7 %in% 3:5 ~ "HS_SomeCol",
  data$educ7 %in% 6:7 ~ "BA_plus",
  TRUE ~ NA_character_
), levels = c("HS_SomeCol", "NoHS", "BA_plus")) # HS = reference

```

1.4 Make year a factor and convert remaining categoricals

```

data$year <- factor(data$year)

cont_vars <- c("age", "dem_therm", "rep_therm")
y_vars <- c("y_dem_like", "y_dem_dislike", "y_rep_like", "y_rep_dislike")
cat_vars <- setdiff(names(data), c(cont_vars, y_vars, "year"))

# already-factor columns pass through unchanged; numeric ones get factored
data[cat_vars] <- lapply(data[cat_vars],
  function(z) if (is.factor(z)) z else factor(z))

```

1.5 Listwise-delete missing predictors

```
data_full <- na.omit(data)
dim(data_full)
```

```
## [1] 17844    25
```

```
# Every outcome should have both 0s and 1s, every year should have substantial counts.
cat("\nClass distribution of each outcome:\n")
```

```
##
## Class distribution of each outcome:
```

```
for (y in y_vars) {
  cat(sprintf("  %-15s ", y)); print(table(data_full[[y]]))
}
```

```
##   y_dem_like
##    0     1
## 9420 8424
##   y_dem_dislike
##    0     1
## 8891 8953
##   y_rep_like
##    0     1
## 10549 7295
##   y_rep_dislike
##    0     1
## 8159 9685
```

```
cat("\nYear counts:\n"); print(table(data_full$year))
```

```
##
## Year counts:
```

```
##
## 2008 2012 2016 2020 2024
## 1251 4456 2618 5772 3747
```

1.6 Build X and Y

```
Y_dl <- data_full$y_dem_like

# RHS = everything except the four outcomes
rhs <- setdiff(names(data_full), y_vars)
form <- as.formula(paste("~", paste(rhs, collapse = " + ")))
X <- model.matrix(form, data = data_full)[, -1] # drop intercept col
dim(X)
```

```
## [1] 17844    50
```

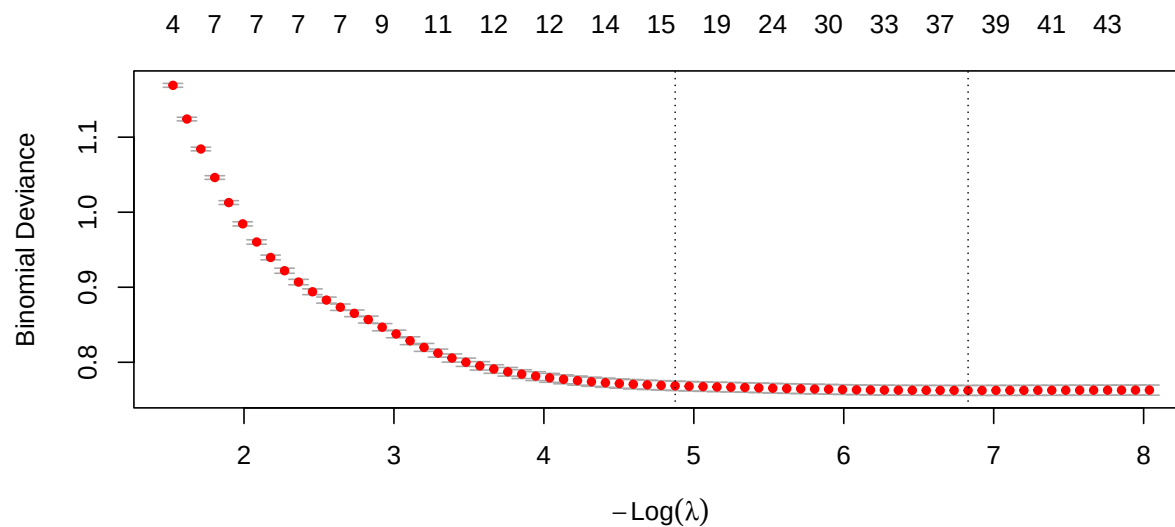
1.7 Force year in via penalty.factor

```
pf <- ifelse(grepl("^year", colnames(X)), 0, 1)
sum(pf == 0) # number of year dummies forced in (= 4 for 5 years)
```

```
## [1] 4
```

1.8 Run cv.glmnet

```
set.seed(10)
fit.dl.cv <- cv.glmnet(X, Y_dl,
                       alpha      = 1,
                       family     = "binomial",
                       nfolds     = 10,
                       penalty.factor = pf)
plot(fit.dl.cv)
```



```
fit.dl.cv$lambda.min
```

```
## [1] 0.00108
```

```
fit.dl.cv$lambda.1se
```

```
## [1] 0.00763
```

1.9 Extract selected variables at lambda.1se

```
coef.dl <- extract.coef(fit.dl.cv, lambda = fit.dl.cv$lambda.1se)
coef.dl
```

##	Value SE	Coefficient
## (Intercept)	-1.6623 NA	(Intercept)
## year2012	-0.5289 NA	year2012
## year2016	-11.0713 NA	year2016
## year2020	-0.4951 NA	year2020
## year2024	-0.2696 NA	year2024
## dem_therm	0.0292 NA	dem_therm
## rep_therm	-0.0159 NA	rep_therm
## pid7Dem	0.5083 NA	pid7Dem
## party_diff2	0.5227 NA	party_diff2
## party_more_consv2	0.0470 NA	party_more_consv2
## party_more_consv9	-0.2349 NA	party_more_consv9
## dem_ideoLib	0.1467 NA	dem_ideoLib
## rep_ideoCon	0.6733 NA	rep_ideoCon
## ideologyLib	0.2140 NA	ideologyLib
## educ7BA_plus	0.4312 NA	educ7BA_plus
## health_insur2	-0.0387 NA	health_insur2

```
var.dl <- rownames(coef.dl)[-1] # column names from model.matrix
```

1.10 Refit with glm (relaxed LASSO) and apply clustered SEs

model.matrix produces names like pid7Rep, race_eth5. We map them back to parent variable names to write the glm formula.

```
parent_var <- function(cols, vars) {
  hits <- sapply(cols, function(cn) {
    m <- vars[startsWith(cn, vars)]
    if (length(m)) m[which.max(nchar(m))] else NA_character_
  })
  unique(na.omit(hits))
}

vars.dl <- parent_var(var.dl, names(data_full))
vars.dl <- union(vars.dl, "year") # year is forced in regardless
```

```
f.dl <- as.formula(paste("y_dem_like ~", paste(vars.dl, collapse = " + ")))
fit.dl <- glm(f.dl, data = data_full, family = binomial)

V.dl <- sandwich::vcovCL(fit.dl, cluster = ~ year, type = "HC1")
coeftest(fit.dl, vcov. = V.dl)
```

```
##
## z test of coefficients:
##
##
```

	Estimate	Std. Error	z value	Pr(> z)
## (Intercept)	-2.52e+00	1.11e-01	-22.76	< 2e-16 ***
## year2012	-5.55e-01	2.25e-02	-24.65	< 2e-16 ***
## year2016	-2.01e+01	1.12e+00	-18.00	< 2e-16 ***
## year2020	-5.52e-01	3.90e-02	-14.14	< 2e-16 ***
## year2024	-2.98e-01	3.74e-02	-7.97	1.6e-15 ***
## dem_therm	3.31e-02	1.25e-03	26.58	< 2e-16 ***
## rep_therm	-1.88e-02	7.73e-04	-24.30	< 2e-16 ***
## pid7Dem	6.96e-01	9.33e-02	7.46	8.9e-14 ***
## pid7Rep	3.19e-01	7.70e-02	4.14	3.4e-05 ***
## party_diff2	5.89e-01	8.06e-02	7.31	2.7e-13 ***
## party_diff9	6.31e-02	5.63e-01	0.11	0.9108
## party_more_consv2	1.37e-01	9.71e-02	1.41	0.1597
## party_more_consv9	-2.45e-01	1.40e-01	-1.74	0.0811 .
## dem_ideoLib	3.89e-01	4.59e-02	8.47	< 2e-16 ***
## dem_ideoCon	3.15e-01	1.10e-01	2.85	0.0043 **
## rep_ideoLib	3.65e-01	1.21e-01	3.02	0.0025 **
## rep_ideoCon	9.27e-01	9.95e-02	9.31	< 2e-16 ***
## ideologyLib	2.86e-01	6.49e-02	4.41	1.1e-05 ***
## ideologyCon	2.40e-02	8.12e-02	0.30	0.7679
## educ7NoHS	-1.51e-01	1.00e-01	-1.51	0.1315
## educ7BA_plus	5.05e-01	1.00e-01	5.04	4.7e-07 ***
## health_insur2	-1.99e-01	8.06e-02	-2.47	0.0133 *

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```


2 Loop over all four outcomes

Same logic, wrapped in a function.

```
run_one <- function(y_name, data_full, seed = 10) {
  Y <- as.numeric(data_full[[y_name]])
  stopifnot(length(unique(Y)) == 2) # fail loud if Y is constant

  rhs <- setdiff(names(data_full), y_vars)
  form <- as.formula(paste("~", paste(rhs, collapse = " + ")))
  X <- model.matrix(form, data = data_full)[, -1]
  pf <- ifelse(grepl("^year", colnames(X)), 0, 1)

  set.seed(seed)
  cv <- cv.glmnet(X, Y, alpha = 1, family = "binomial",
                 nfolds = 10, penalty.factor = pf)

  coef.tab <- extract.coef(cv, lambda = cv$lambda.1se)
  vars <- parent_var(rownames(coef.tab)[-1], names(data_full))
  vars <- union(vars, "year")

  f <- as.formula(paste(y_name, "~", paste(vars, collapse = " + ")))
  fit <- glm(f, data = data_full, family = binomial)
  V <- sandwich::vcovCL(fit, cluster = ~ year, type = "HC1")

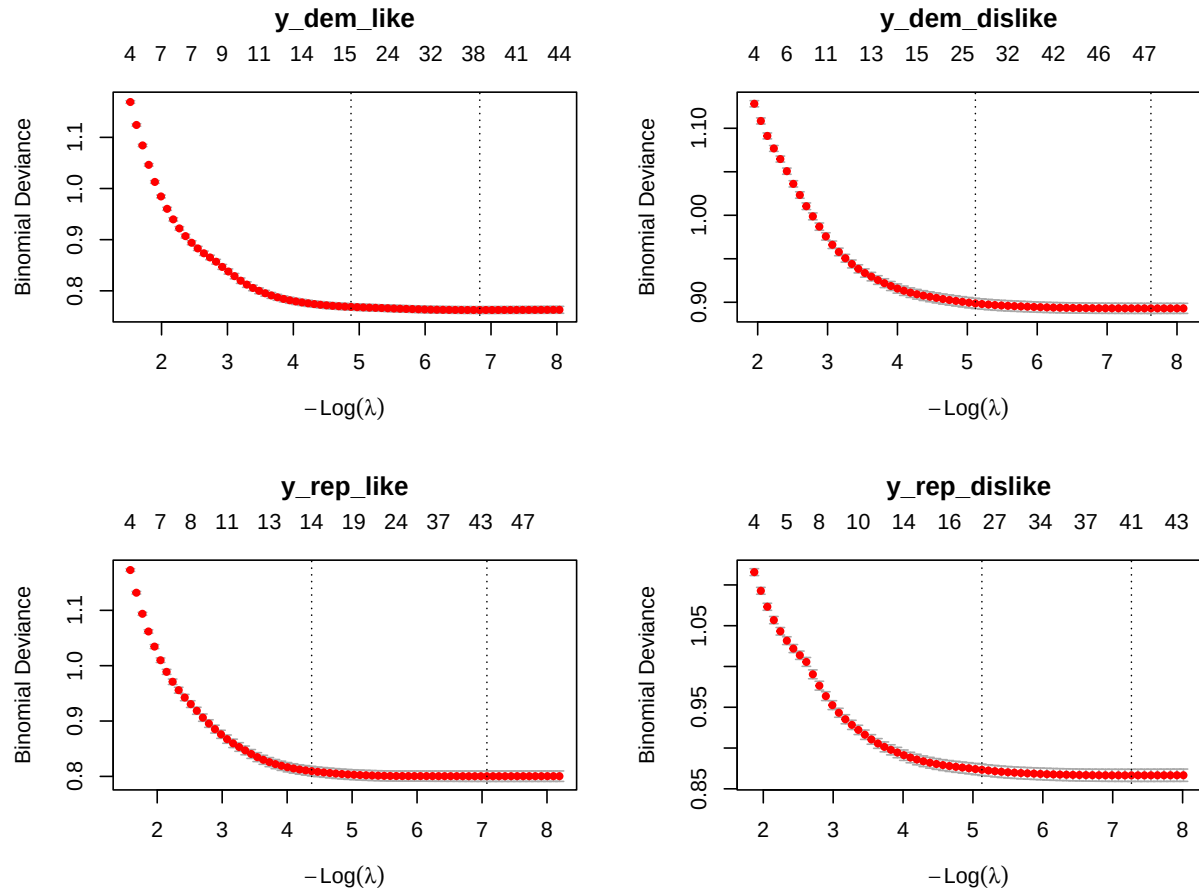
  list(cv = cv, fit = fit, vcov = V,
       coef_test = coeftest(fit, vcov. = V),
       n = nrow(data_full),
       lambda_1se = cv$lambda.1se,
       n_selected = nrow(coef.tab) - 1)
}

results <- lapply(y_vars, run_one, data_full = data_full)
names(results) <- y_vars

data.frame(outcome = y_vars,
           n = sapply(results, `[`, "n"),
           lambda_1se = sapply(results, `[`, "lambda_1se"),
           n_selected = sapply(results, `[`, "n_selected"))
```

```
##               outcome      n lambda_1se n_selected
## y_dem_like      y_dem_like 17844    0.00763      15
## y_dem_dislike y_dem_dislike 17844    0.00601      26
## y_rep_like      y_rep_like  17844    0.01256      14
## y_rep_dislike y_rep_dislike 17844    0.00593      25
```

```
par(mfrow = c(2, 2))
for (y in y_vars) { plot(results[[y]]$cv); title(y, line = 2.5) }
```



```
par(mfrow = c(1, 1))
```

```
for (y in y_vars) {
  cat("\n----- ", y, " ----- \n")
  print(results[[y]]$coef_test)
}
```

```
##
## ----- y_dem_like -----
##
## z test of coefficients:
##
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -2.52e+00  1.11e-01 -22.76 < 2e-16 ***
## year2012      -5.55e-01  2.25e-02 -24.65 < 2e-16 ***
## year2016      -2.01e+01  1.12e+00 -18.00 < 2e-16 ***
## year2020      -5.52e-01  3.90e-02 -14.14 < 2e-16 ***
## year2024      -2.98e-01  3.74e-02  -7.97 1.6e-15 ***
## dem_therm       3.31e-02  1.25e-03  26.58 < 2e-16 ***
## rep_therm      -1.88e-02  7.73e-04 -24.30 < 2e-16 ***
## pid7Dem         6.96e-01  9.33e-02   7.46 8.9e-14 ***
## pid7Rep         3.19e-01  7.70e-02   4.14 3.4e-05 ***
## party_diff2     5.89e-01  8.06e-02   7.31 2.7e-13 ***
```

```

## party_diff9      6.31e-02  5.63e-01  0.11  0.9108
## party_more_consv2 1.37e-01  9.71e-02  1.41  0.1597
## party_more_consv9 -2.45e-01  1.40e-01 -1.74  0.0811 .
## dem_ideoLib      3.89e-01  4.59e-02  8.47 < 2e-16 ***
## dem_ideoCon      3.15e-01  1.10e-01  2.85  0.0043 **
## rep_ideoLib      3.65e-01  1.21e-01  3.02  0.0025 **
## rep_ideoCon      9.27e-01  9.95e-02  9.31 < 2e-16 ***
## ideologyLib      2.86e-01  6.49e-02  4.41  1.1e-05 ***
## ideologyCon      2.40e-02  8.12e-02  0.30  0.7679
## educ7NoHS       -1.51e-01  1.00e-01 -1.51  0.1315
## educ7BA_plus     5.05e-01  1.00e-01  5.04  4.7e-07 ***
## health_insur2    -1.99e-01  8.06e-02 -2.47  0.0133 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## ----- y_dem_dislike -----
##
## z test of coefficients:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -2.20e+00  2.44e-01  -9.01 < 2e-16 ***
## year2012       5.15e-01  1.50e-02  34.26 < 2e-16 ***
## year2016     -1.90e+01  1.12e+00 -16.96 < 2e-16 ***
## year2020       6.11e-01  3.72e-02  16.44 < 2e-16 ***
## year2024       6.99e-01  2.35e-02  29.72 < 2e-16 ***
## reg_census2   -1.58e-01  7.96e-02  -1.99  0.04635 *
## reg_census3   -3.17e-02  4.45e-02  -0.71  0.47590
## reg_census4    1.45e-01  1.86e-02   7.78  7.4e-15 ***
## dem_therm     -2.60e-02  2.23e-03 -11.66 < 2e-16 ***
## pid7Dem       3.09e-01  1.06e-01   2.91  0.00356 **
## pid7Rep       3.71e-01  8.46e-02   4.39  1.1e-05 ***
## party_diff2    5.70e-01  5.09e-02  11.20 < 2e-16 ***
## party_diff9    4.35e-01  6.24e-01   0.70  0.48619
## party_more_consv2 6.37e-01  1.05e-01   6.06  1.4e-09 ***
## party_more_consv9 1.71e-01  9.77e-02   1.75  0.07949 .
## dem_ideoLib    3.13e-01  5.77e-02   5.43  5.8e-08 ***
## dem_ideoCon    1.81e-01  4.47e-02   4.05  5.1e-05 ***
## rep_ideoLib    2.28e-01  5.57e-02   4.10  4.1e-05 ***
## rep_ideoCon    8.37e-01  1.10e-01   7.57  3.6e-14 ***
## right_track2   1.91e-01  7.30e-02   2.62  0.00875 **
## ideologyLib    2.04e-01  5.59e-02   3.65  0.00026 ***
## ideologyCon    4.76e-01  8.19e-02   5.81  6.1e-09 ***
## age           1.31e-02  8.04e-04  16.30 < 2e-16 ***
## gender2       -2.87e-01  7.06e-02  -4.07  4.8e-05 ***
## gender3        9.11e-01  4.60e-02  19.78 < 2e-16 ***
## race_eth2     -1.38e-01  3.75e-02  -3.67  0.00024 ***
## race_eth3     -4.38e-01  4.01e-02 -10.91 < 2e-16 ***
## race_eth4     -9.53e-02  2.81e-01  -0.34  0.73418
## race_eth5     -2.03e-01  1.45e-02 -14.01 < 2e-16 ***
## race_eth6      1.83e-01  3.64e-02   5.03  4.8e-07 ***
## work_status2   -5.08e-02  9.42e-02  -0.54  0.58936
## work_status3   -1.31e-01  5.33e-02  -2.46  0.01381 *
## work_status4   -7.95e-02  1.18e-01  -0.67  0.50155

```

```

## work_status5      4.24e-01  1.14e-01   3.71  0.00021 ***
## educ7NoHS        -4.35e-01  6.13e-02  -7.10  1.3e-12 ***
## educ7BA_plus     6.28e-01  2.81e-02  22.35 < 2e-16 ***
## marital_status2  -5.28e-02  1.28e-02  -4.13  3.7e-05 ***
## marital_status3  -2.11e-01  4.91e-02  -4.31  1.7e-05 ***
## marital_status4   1.72e-01  8.07e-02   2.14  0.03261 *
## marital_status5   9.43e-03  4.67e-02   0.20  0.84003
## marital_status7  -5.04e-02  4.89e-02  -1.03  0.30286
## health_insurr2   -1.34e-01  1.08e-01  -1.24  0.21551
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## ----- y_rep_like -----
##
## z test of coefficients:
##
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -3.46042   0.09771  -35.42 < 2e-16 ***
## year2012         0.24247   0.01379   17.58 < 2e-16 ***
## year2016    -19.04516   1.12275  -16.96 < 2e-16 ***
## year2020         0.22564   0.03412    6.61 3.7e-11 ***
## year2024         0.38695   0.03217   12.03 < 2e-16 ***
## dem_therm       -0.01400   0.00125  -11.16 < 2e-16 ***
## rep_therm        0.02589   0.00146   17.79 < 2e-16 ***
## pid7Dem          0.12174   0.08089    1.51  0.1323
## pid7Rep          0.57777   0.03645   15.85 < 2e-16 ***
## party_diff2       0.63872   0.09430    6.77 1.3e-11 ***
## party_diff9       0.76400   0.11722    6.52 7.1e-11 ***
## party_more_consv2 0.45665   0.10446    4.37 1.2e-05 ***
## party_more_consv9 0.08950   0.12967    0.69  0.4900
## dem_ideoLib       0.70227   0.03127   22.46 < 2e-16 ***
## dem_ideoCon       0.16488   0.14041    1.17  0.2403
## rep_ideoLib       0.21844   0.13196    1.66  0.0979 .
## rep_ideoCon       0.76167   0.07581   10.05 < 2e-16 ***
## ideologyLib      -0.17706   0.05701   -3.11  0.0019 **
## ideologyCon       0.41484   0.06184    6.71 2.0e-11 ***
## gender2          -0.19177   0.02398   -8.00 1.3e-15 ***
## gender3           1.61857   0.10824   14.95 < 2e-16 ***
## educ7NoHS        -0.22006   0.03889   -5.66 1.5e-08 ***
## educ7BA_plus     0.55097   0.06675    8.25 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## ----- y_rep_dislike -----
##
## z test of coefficients:
##
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -0.85928   0.34916   -2.46 0.01385 *
## year2012        -0.52420   0.04217  -12.43 < 2e-16 ***
## year2016    -20.32189   1.12170  -18.12 < 2e-16 ***
## year2020        -0.69123   0.05819  -11.88 < 2e-16 ***

```

```

## year2024          -0.41554    0.06725    -6.18    6.4e-10 ***
## reg_census2        0.02001    0.02926     0.68    0.49408
## reg_census3       -0.03611    0.03170    -1.14    0.25458
## reg_census4        0.18802    0.05679     3.31    0.00093 ***
## dem_therm         0.00409    0.00112     3.64    0.00027 ***
## rep_therm        -0.03316    0.00245   -13.51    < 2e-16 ***
## pid7Dem           0.35278    0.15920     2.22    0.02669 *
## pid7Rep            0.27400    0.06032     4.54    5.6e-06 ***
## party_diff2        0.67958    0.11595     5.86    4.6e-09 ***
## party_diff9        0.33354    0.41915     0.80    0.42617
## party_more_consv2  0.61600    0.06084    10.13    < 2e-16 ***
## party_more_consv9  0.03637    0.06819     0.53    0.59379
## dem_ideoLib        0.57275    0.02381    24.05    < 2e-16 ***
## dem_ideoCon        0.37481    0.14942     2.51    0.01213 *
## rep_ideoLib        0.18409    0.01672    11.01    < 2e-16 ***
## rep_ideoCon        0.50985    0.02862    17.82    < 2e-16 ***
## ideologyLib        0.23297    0.06558     3.55    0.00038 ***
## ideologyCon        0.27844    0.01768    15.75    < 2e-16 ***
## age                0.01561    0.00239     6.55    5.9e-11 ***
## gender2           -0.29923    0.05996    -4.99    6.0e-07 ***
## gender3           -0.28945    0.02825   -10.25    < 2e-16 ***
## race_eth2         -0.16098    0.07404    -2.17    0.02970 *
## race_eth3         -0.54668    0.13347    -4.10    4.2e-05 ***
## race_eth4          0.28030    0.25907     1.08    0.27927
## race_eth5         -0.20714    0.07870    -2.63    0.00849 **
## race_eth6          0.08594    0.08385     1.02    0.30539
## relg2             -0.01751    0.05997    -0.29    0.77032
## relg3             -0.01278    0.09043    -0.14    0.88763
## relg4              0.13519    0.03031     4.46    8.2e-06 ***
## educ7NoHS         -0.48186    0.04680   -10.30    < 2e-16 ***
## educ7BA_plus       0.60313    0.04253    14.18    < 2e-16 ***
## marital_status2   -0.07989    0.05479    -1.46    0.14478
## marital_status3   -0.19654    0.00248   -79.26    < 2e-16 ***
## marital_status4   -0.04857    0.22515    -0.22    0.82918
## marital_status5   -0.21383    0.04880    -4.38    1.2e-05 ***
## marital_status7   -0.16615    0.04665    -3.56    0.00037 ***
## health_insur2     -0.08872    0.15920    -0.56    0.57736
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

3 Robustness check: pooled outcome

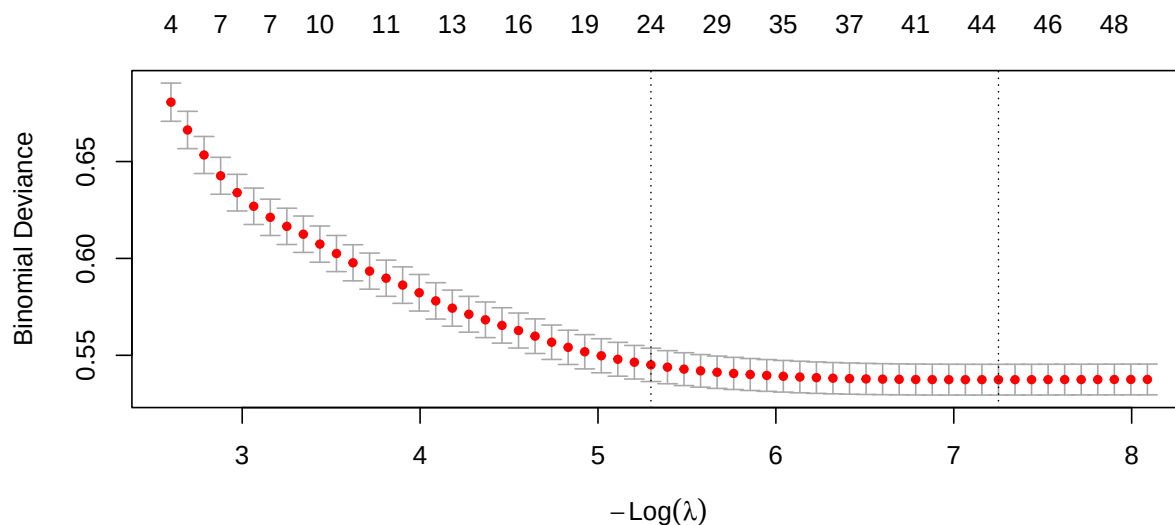
A respondent is **in the open-ended sample** if they provided usable text on *any* of the four items. One binary outcome instead of four; if the same predictors fall out we have evidence the per-item models aren't picking up item-specific noise.

```
data_full$y_pooled <- as.integer(
  data_full$y_dem_like == 1 |
  data_full$y_dem_dislike == 1 |
  data_full$y_rep_like == 1 |
  data_full$y_rep_dislike == 1
)
table(data_full$y_pooled, useNA = "ifany")
```

```
##
##      0      1
## 4741 13103
```

```
Y_p <- data_full$y_pooled
rhs_p <- setdiff(names(data_full), c(y_vars, "y_pooled"))
form_p <- as.formula(paste("~", paste(rhs_p, collapse = " + ")))
X_p <- model.matrix(form_p, data = data_full)[, -1]
pf_p <- ifelse(grepl("^year", colnames(X_p)), 0, 1)

set.seed(10)
fit.p.cv <- cv.glmnet(X_p, Y_p, alpha = 1, family = "binomial",
  nfolds = 10, penalty.factor = pf_p)
plot(fit.p.cv)
```



```
coef.p <- extract.coef(fit.p.cv, lambda = fit.p.cv$lambda.1se)
coef.p
```

##	Value	SE	Coefficient
## (Intercept)	-1.58e+00	NA	(Intercept)
## year2012	-3.35e-01	NA	year2012
## year2016	-1.27e+01	NA	year2016
## year2020	-2.07e-01	NA	year2020
## year2024	-3.54e-02	NA	year2024
## rep_therm	-6.00e-03	NA	rep_therm
## pid7Dem	7.67e-01	NA	pid7Dem
## pid7Rep	5.48e-01	NA	pid7Rep
## party_diff2	9.02e-01	NA	party_diff2
## party_more_consv2	4.06e-01	NA	party_more_consv2
## party_more_consv9	-1.14e-01	NA	party_more_consv9
## dem_ideoLib	3.69e-01	NA	dem_ideoLib
## dem_ideoCon	1.64e-01	NA	dem_ideoCon
## rep_ideoLib	4.77e-02	NA	rep_ideoLib
## rep_ideoCon	8.20e-01	NA	rep_ideoCon
## right_track1	1.30e-02	NA	right_track1
## right_track2	-2.75e-16	NA	right_track2
## ideologyLib	2.47e-01	NA	ideologyLib
## ideologyCon	5.71e-01	NA	ideologyCon
## age	2.03e-02	NA	age
## gender2	-1.96e-01	NA	gender2
## race_eth3	-1.20e-01	NA	race_eth3
## educ7NoHS	-3.57e-02	NA	educ7NoHS
## educ7BA_plus	4.49e-01	NA	educ7BA_plus
## health_insur2	-7.32e-02	NA	health_insur2

```
vars.p <- parent_var(rownames(coef.p)[-1], names(data_full))
vars.p <- union(vars.p, "year")

f.p <- as.formula(paste("y_pooled ~", paste(vars.p, collapse = " + ")))
fit.p <- glm(f.p, data = data_full, family = binomial)
V.p <- sandwich::vcovCL(fit.p, cluster = ~ year, type = "HC1")
coeftest(fit.p, vcov. = V.p)
```

```
##
## z test of coefficients:
##
##          Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -2.18e+00   2.14e-01  -10.22 < 2e-16 ***
## year2012      -4.01e-01   4.99e-02   -8.04 9.3e-16 ***
## year2016      -2.14e+01   1.13e+00  -19.03 < 2e-16 ***
## year2020      -2.97e-01   7.37e-02   -4.03 5.6e-05 ***
## year2024      -1.30e-01   6.49e-02   -2.00 0.0457 *
## rep_therm     -1.13e-02   8.35e-04  -13.49 < 2e-16 ***
## pid7Dem        1.10e+00   8.99e-02   12.27 < 2e-16 ***
## pid7Rep        1.11e+00   8.95e-02   12.36 < 2e-16 ***
## party_diff2     9.15e-01   4.66e-02   19.64 < 2e-16 ***
## party_diff9     3.30e-01   1.01e-01    3.27 0.0011 **
## party_more_consv2 4.94e-01   7.23e-02    6.83 8.4e-12 ***
## party_more_consv9 -4.78e-02   1.62e-01   -0.30 0.7674
## dem_ideoLib     5.24e-01   5.60e-02    9.35 < 2e-16 ***
## dem_ideoCon     4.72e-01   6.19e-02    7.64 2.2e-14 ***
## rep_ideoLib     3.29e-01   1.41e-01    2.34 0.0192 *
```

```

## rep_ideoCon      9.98e-01  1.39e-01   7.16  7.9e-13 ***
## right_track2     -1.56e-01  5.99e-02  -2.60  0.0094 **
## ideologyLib      4.17e-01  8.28e-02   5.03  4.8e-07 ***
## ideologyCon      7.40e-01  5.35e-02  13.83 < 2e-16 ***
## age              2.52e-02  2.93e-03   8.59 < 2e-16 ***
## gender2          -3.40e-01  5.23e-02  -6.51  7.5e-11 ***
## gender3           4.98e-01  1.04e-01   4.79  1.7e-06 ***
## race_eth2         1.94e-01  1.08e-01   1.80  0.0724 .
## race_eth3        -4.38e-01  1.47e-01  -2.98  0.0029 **
## race_eth4         4.66e-01  1.76e-01   2.65  0.0081 **
## race_eth5        -5.58e-03  1.06e-01  -0.05  0.9579
## race_eth6         1.88e-01  9.27e-02   2.03  0.0426 *
## educ7NoHS        -1.82e-01  1.67e-01  -1.09  0.2753
## educ7BA_plus      5.58e-01  7.69e-02   7.26  3.9e-13 ***
## health_insur2    -2.21e-01  1.34e-01  -1.65  0.0994 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```


4 Summary

Across all four per-item models and the pooled robustness check, four predictor families do most of the work.

`y_dem_like` is the simplest model (15 predictors): partisanship, ideology placement, thermometers, and education explain who articulates Democratic likes.

`y_dem_dislike` is the richest model (26 predictors), pulling in age, race, gender, marital status, and work status. Articulating dislike of Democrats appears to be more demographically structured than articulating likes.

`y_rep_like` (14 predictors) is dominated by partisanship and ideology — demographics drop out almost entirely. Whether someone can articulate Republican likes is essentially a function of political engagement.

`y_rep_dislike` (25 predictors) again brings in demographics, similar to `y_dem_dislike`.

Below are some general findings -

1. Education is the single most consistent variable. `educ7BA_plus` is positive and highly significant in all four per-item models and the pooled model, and `educ7NoHS` is consistently negative. Holding everything else constant, college-educated respondents are roughly 60% more likely (in odds-ratio terms) to provide usable open-ended text than the high-school-or-some-college reference group.
2. Partisan strength matters more than direction. Both `pid7Dem` and `pid7Rep` show positive coefficients in most models, indicating that identifying with either party (versus pure independents, the reference) predicts response.
3. Political knowledge / engagement variables are extremely strong. `party_diff2` (“yes there’s a difference between the parties”) has $\beta = 0.55\text{--}0.92$ in every model — one of the largest selected effects. `party_more_consv2` (correctly identifying Republicans as more conservative) is also consistently positive.

Beyond education, demographics matter most for the dislike outcomes:

1. Age is positive and significant for both dislike outcomes ($\beta = 0.013\text{--}0.016$ per year): older respondents are more likely to articulate dislikes. It does not appear in either like model at lambda lse.
2. Gender is selected in three of four models. Women (`gender2`) are slightly less likely to give open-ended responses than men; the “other” category (`gender3`) shows large effects in opposite directions across items, but the small cell size warrants caution.
3. Race/ethnicity is selected for `y_dem_dislike` and `y_rep_dislike`. Asian (`race_eth3`) and Hispanic (`race_eth5`) respondents are consistently less likely to give open-ended responses than the white-non-Hispanic reference, all else equal.